

Emotional Intelligence & Intent Determination Stress Benchmark Report

Date: 2026-08-21
Evaluation Target: Single Dedicated GPU (NVIDIA RTX A6000 - GPU 2, `CUDA_VISIBLE_DEVICES="2"`)
Evaluated Stack: Emotional Tone Classifier (`app.tools.empathy`) + Supervisor Intent Router (`app.agents.supervisor`) + Multi-Turn Frontend Chat API (`/v1/chat`)
Evaluated Dimensions: Emotion Distress Classification, Intensity Scoring, Empathetic Acknowledgement, Multilingual Intent Determination (EN, LG, SW), and Concurrent Stress (\$c = 10 \to 250\$).

1. Executive Summary

This report provides comprehensive empirical verification of the **Emotional Intelligence (EI)** and **Intent Determination** systems in the URA Tax Assistant across English, Luganda, and Swahili under concurrent load, stress, spike, and volume testing.

Key Highlights

- **Emotional Intelligence Recognition: 100.0% accuracy** (6/6) across distress categories (`hardship` , `frustration` , `anxiety` , `urgency` , `confusion` , `neutral`) with sub-millisecond execution (`0.053 – 0.120 ms`).
- **Intensity & De-escalation Calibration:** Accurately distinguished between mild confusion and extreme panicking/ hardship, triggering appropriate human handoff recommendations and empathetic conversational openers.
- **Multilingual Intent Determination: 100.0% precision** across tax calculations, FAQ policy lookups, formal objection/appeal filings, human officer escalations, and conversational courtesy across English, Luganda, and Swahili.
- **Concurrency Scaling Curve (\$c = 10 \to 100\$):** Maintained **100.0% accuracy** and zero errors across all concurrency tiers.
- **Extreme Concurrency Stress (\$c = 250\$, 500 requests): 100.0% response accuracy** at **41.7 RPS** (p50: 5,198.8 ms).
- **Traffic Spike Surge (\$c = 250\$ in 50ms):** Processed 250 sudden burst queries at **46.7 RPS** with **0 circuit breaker trips**.
- **High-Volume Soak (1,500 Continuous Queries):** Completed 1,500 queries in **55.93s (26.8 RPS)** with **0.0 MiB** memory leak.
- **Hardware Isolation & Cleanup:** Dedicated execution on GPU 2; process cleanup strictly scoped to `~/Mpairwe7` with zero interference with external cluster jobs.

2. Emotional Intelligence (EI) & Distress Recognition Matrix

Test ID	Distress Category	Sample Taxpayer Input	Detected Kind	Intensity	Human Handoff	Empathetic Opener Generated	Latency (ms)	Status
EMO-01	Hardship	"I am completely broke and bankrupt, I cannot afford this tax and I will lose my business and shop!"	hardship	high	TRUE	"I'm sorry you're going through this — let's look at what options you have."	0.120 ms	PASS ✓
EMO-02	Frustration	"EFRIS has failed again for the fourth time today!! Nothing is working and I am EXTREMELY angry!"	frustration	high	TRUE	"I'm sorry for the trouble — let's get this sorted out together."	0.084 ms	PASS ✓
EMO-03	Anxiety	"I am so worried and scared because my bank accounts have been frozen by an agency notice and I am panicking."	anxiety	moderate	false	"I understand this can feel stressful — let's take it step by step."	0.074 ms	PASS ✓
EMO-04	Urgency	"URGENT: I need to submit my return before midnight today or I will be fined!"	urgency	moderate	false	"I know deadlines can be stressful, so here's the quickest way forward."	0.073 ms	PASS ✓

Test ID	Distress Category	Sample Taxpayer Input	Detected Kind	Intensity	Human Handoff	Empathetic Opener Generated	Latency (ms)	Status
EMO-05	Confusion	"I don't understand how gross rental income vs allowable deduction works, it's very complicated and unclear."	confusion	moderate	false	"Let me put that a different way."	0.053 ms	PASS
EMO-06	Neutral	"What is the standard VAT rate in Uganda for FY2026-27?"	none	none	false	Direct answer (no unprompted empathy opener)	0.055 ms	PASS

3. Multilingual Intent Determination Accuracy (EN, LG, SW)

Intent ID	Language	Classified Intent	Sample Taxpayer Query	Supervisor Routing Target	Latency (ms)	Status
INT-EN-01	en	Tax Calculation	"How much PAYE tax should I deduct for gross monthly salary of UGX 4,500,000?"	calculator (calculate_paye)	216.97 ms	PASS
INT-EN-02	en	FAQ Policy Lookup	"What is the threshold for mandatory VAT registration in Uganda?"	rag / FAQ Corpus	55.02 ms	PASS
INT-EN-03	en	Dispute / Objection	"I want to object and dispute an unfair tax assessment issued on my company."	escalate (Objection review)	19.70 ms	PASS
INT-EN-04	en	Human Escalation	"I need to speak directly with a human URA officer right now."	escalate (Staff Ticket Queue)	1,263.47 ms	PASS
INT-EN-05	en	Chitchat / Courtesy	"Good morning! Thank you for your assistance today."	chitchat / Courtesy Handler	649.05 ms	PASS
INT-LG-01	lg	Tax Calculation (Luganda)	"Nnyinza ntya okubala omusolo gwa PAYE ku musaala gwa shs 4,500,000 buli mwezi?"	calculator / Hybrid RAG	754.54 ms	PASS
INT-LG-02	lg	FAQ Lookup (Luganda)	"Biki ebyetaagisa okufuna namba ya TIN mu URA?"	rag / Luganda FAQ Corpus	733.38 ms	PASS

Intent ID	Language	Classified Intent	Sample Taxpayer Query	Supervisor Routing Target	Latency (ms)	Status
INT-LG-03	lg	Dispute Appeal (Luganda)	"Njagala kwongera kwemulugunya ku musolo gwe banzisizza ogutalina nsonga."	rag / Objection Guidance	683.84 ms	PASS
INT-LG-04	lg	Human Escalation (Luganda)	"Njagala kwogera n'omukozi wa URA mu ofiisi amangu ddala."	escalate / Staff Ticket Queue	658.73 ms	PASS
INT-SW-01	sw	Tax Calculation (Swahili)	"Ni kiasi gani cha kodi ya PAYE kinacholipwa kwa mshahara wa UGX 4,500,000?"	calculator / Hybrid RAG	712.18 ms	PASS
INT-SW-02	sw	FAQ Lookup (Swahili)	"Kiwango cha chini cha usajili wa lazima wa VAT ni kiasi gani?"	rag / Swahili FAQ Corpus	691.45 ms	PASS
INT-SW-03	sw	Human Escalation (Swahili)	"Nahitaji kuongea na afisa wa kibinadamu wa mamlaka ya mapato URA mara moja."	escalate / Staff Ticket Queue	665.20 ms	PASS

4. Concurrency Scaling Matrix (\$c = 10 \to 100\$)

Concurrency (\$c\$)	Total Requests	Duration (s)	Throughput (RPS)	Accuracy (%)	Latency p50 (ms)	Latency p95 (ms)	Error Rate (%)	Status
\$c = 10\$	40	1.309s	30.6 RPS	100.0%	316.5 ms	569.2 ms	0.0%	Optimal SLA
\$c = 25\$	50	1.343s	37.2 RPS	100.0%	483.9 ms	864.2 ms	0.0%	Linear Scaling
\$c = 50\$	100	2.766s	36.2 RPS	100.0%	1,008.2 ms	1,847.6 ms	0.0%	Medium Load
\$c = 100\$	200	5.378s	37.2 RPS	100.0%	2,246.3 ms	3,467.1 ms	0.0%	High Load

5. Extreme Stress, Traffic Surge Spike & Volume Soak

A. Extreme Concurrency Stress (\$c = 250\$, 500 Requests)

- **Throughput:** 41.7 RPS (500 requests completed in 11.99s).
- **Classification Accuracy:** 100.0% (0 errors).
- **Latency:** p50 = 5,198.8 ms, p95 = 6,712.4 ms.

B. Instantaneous Traffic Spike Surge (\$c = 250\$ in 50ms)

- **Throughput:** 46.7 RPS (250 requests completed in 5.35s).
- **Stability:** 100.0% accuracy with 0 circuit breaker trips.

- **Latency:** p50 = 4,028.9 ms, p95 = 4,689.3 ms.

C. Sustained High-Volume Soak (1,500 Continuous Queries)

- **Total Workload:** 1,500 continuous multilingual emotional/intent queries under \$c = 50\$.
- **Throughput:** 26.8 RPS (completed in 55.93s).
- **Accuracy Stability:** 100.0% accuracy maintained over all 1,500 requests.
- **Resource Footprint:** 0.0 MiB heap or VRAM leak.

6. Single-GPU Hardware Telemetry & Scoped Cleanup

Metric	Measured Value	Operational Safety Verification
Target GPU	NVIDIA RTX A6000 (GPU 2)	CUDA_VISIBLE_DEVICES="2"
Total Hardware Capacity	49,140 MiB (48.0 GiB)	10,752 CUDA Cores
Active Peak VRAM Consumption	5,905 MiB (12.01%)	42,770 MiB (87.99%) free headroom
Peak GPU Temperature & Power	62.0 °C / 101.0 W	Well below thermal limit and 300W TDP
Post-Test Resource Cleanup	CLEAN (Scoped strictly to ~/Mpairwe7)	0 lingering processes; external jobs untouched

7. Artifacts & Source References

- **Benchmark Runner:** scripts/test_emotional_intelligence_intent_stress.py
- **Scoped Cleanup Tool:** scripts/cleanup_gpu_processes.py
- **Raw Metrics JSON:** Results/metrics/emotional_intelligence_intent_report.json
- **Linked GitHub Issue:** [Issue #304](#)